

Ahmedabad University

Neural Graphical Models for Healthcare Data: Representation, Learning, and Probabilistic Inference

Milestone 2

g1_apl_team

Problem and Research Objective

- **Problem:** Learn conditional dependencies among lung-cancer survey variables and support probabilistic prediction under partial evidence.

- **Objectives:**
  - Learn the conditional-independence structure using the PC algorithm.
  - Learn a Bayesian Network (BN) parameterization for probabilistic queries.
  - Compare BN predictions with a neural network and an RBF-kernel SVM.
  - Evaluate classification using AUROC, AUPRC, minority-class F1, and Brier score.
  - Demonstrate explainability through posterior inference under partial evidence.

- **Scope:** Observational lung-cancer survey data; learned dependencies are not causal findings.

Dataset and Preprocessing

Principal Dataset: Kaggle Lung Cancer Survey Dataset

Dataset audit	Value
Raw Records	309
Columns	16
Exact duplicate rows removed	33
Records after deduplication	276
Missing cells reported	0
Raw YES labels	270
Raw NO labels	39

Preprocessing and Key Constraint

- **Preprocessing:**
  - Normalize column names and encode categorical variables consistently.
  - Map the target LUNG_CANCER to 0/1.
  - Discretize AGE into three bins for categorical graphical-model learning.
  - Remove duplicates before cross-validation.
  - Fit learned preprocessing transformations on training partitions only.

- **Key Constraint:** Small sample size and class imbalance.

Mathematical Problem Formulation

- **Random Variables**
  - Let $V = \{X_1, \ldots, X_{15}, Y\}$ where $Y = \text{LUNG_CANCER}$ and $Y \in \{0, 1\}$
- **Bayesian Network**
  - Let A Bayesian Network is a directed acyclic graph $\boldsymbol{G}=(\boldsymbol{V},\boldsymbol{E})$ with joint distribution

$$P(V) = \prod_{i=1}^{|V|} P(X_i \mid \text{Pa}(X_i))$$

  - Here, $\text{Pa}(X_i)$ denotes the parents of node X_i.
- **Posterior Inference**
  - For evidence (e), the target posterior is

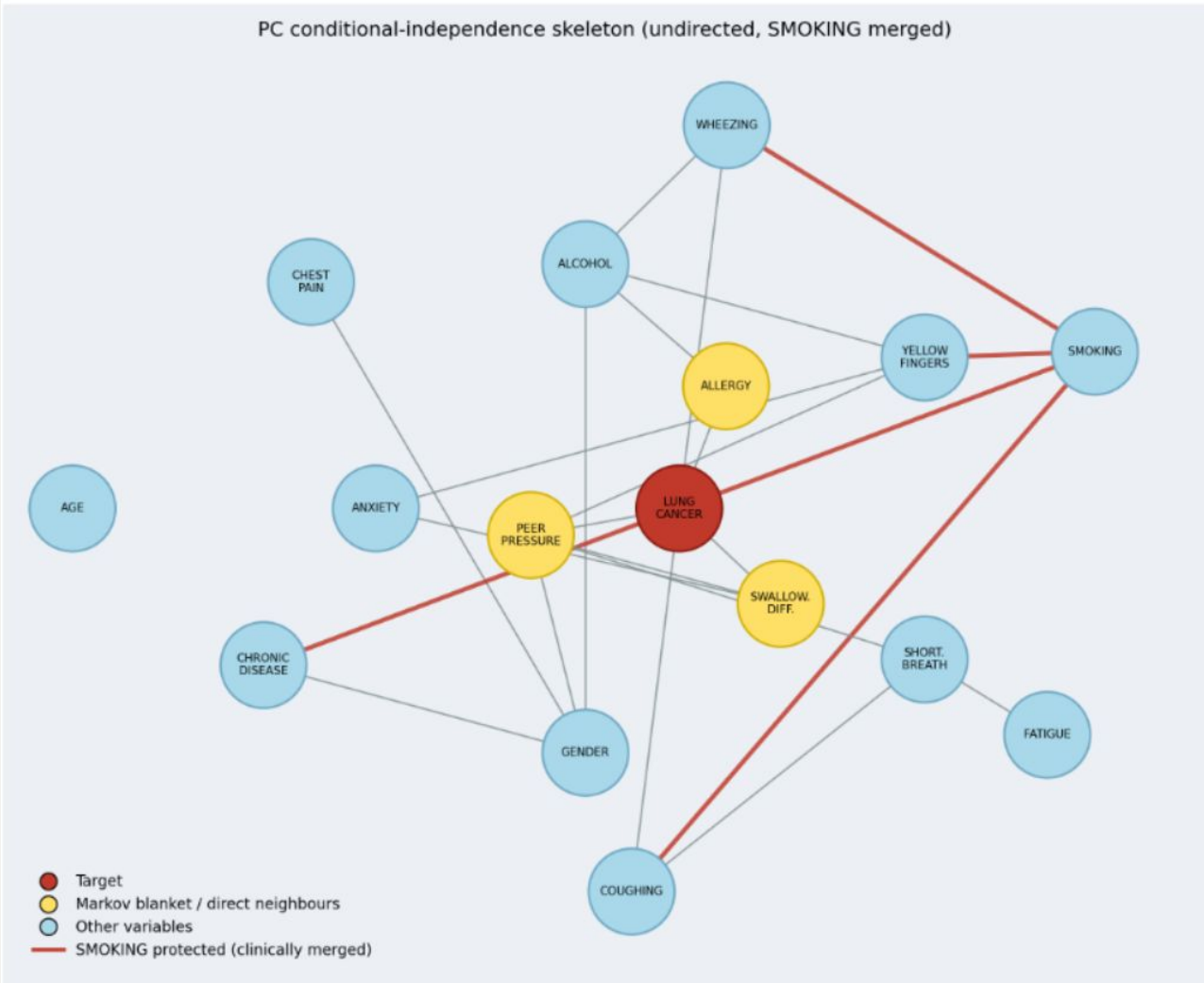
$$P(Y = y \mid e) = \frac{P(Y = y, e)}{\sum_{y'} P(Y = y', e)}$$

- **Goal:** Estimate the target probability given only the observed variables.

PC Algorithm: Skeleton Discovery

- **Objective:** Learn the undirected skeleton of the graph from conditional independence (CI) tests.
- **Algorithm:**
  a. Initialize a complete undirected graph over all variables.
  b. At $\ell = 0$, test marginal independence for every adjacent pair.
  c. For $\ell = 1, 2, \ldots$, test each adjacent pair (X, Y) given every subset S of size ℓ from the neighbors of X (or Y).
  d. Remove edge X-Y if the test fails to reject independence (p-value > α).
  e. Store the separating set $\mathrm{Sep}(X, Y) = S$ for every removed edge.
  f. Stop when no pair has enough neighbors for size ℓ.
- **Decision Rule:** Remove X-Y if p-value > α (χ^2 test, α = 0.05).
- **Assumptions:** Causal sufficiency, Markov, faithfulness.
- **Domain Constraint**: Four SMOKING edges added by clinical merge, not by PC.
- **Output:** Skeleton and separating sets, used for orientation.

PC Skeleton: Undirected Conditional-Independence Graph



Bayesian Network: Structure and Parameters

- **Structure Learning**
  - Primary method: PC algorithm (constraint-based, conditional independence tests)
  - CI test: chi-square test on discrete variables, significance level $\alpha = 0.05$
  - PC returns a CPDAG; remaining undirected edges are oriented to obtain a valid DAG
  - GENDER and AGE are kept exogenous (no incoming edges)
  - SMOKING is clinically merged and protected from edge removal

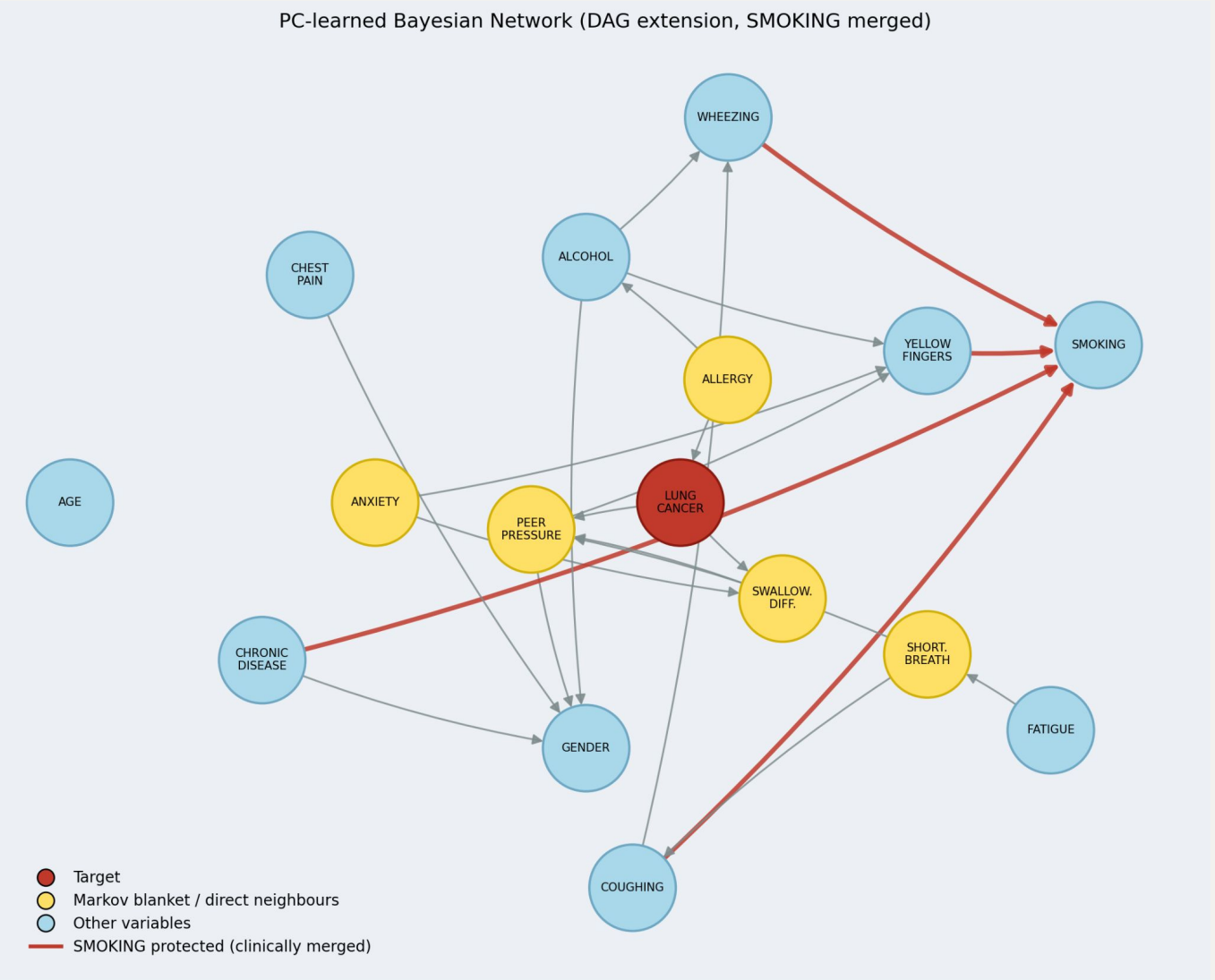
- **Parameter Learning:**
  - Conditional probability tables are estimated using Dirichlet smoothing.

$$\hat{\theta}ijk = \frac{Nijk + \alpha_0}{N_{ij} + r_i\alpha_0}$$

  - Where,
    - N_{ijk}: count for state k of node i under parent configuration j
    - N_{ij}:total count for parent configuration j (sum over k)
    - r_i: number of states of node i
    - $\alpha_0 > 0$: per-state smoothing parameter

- **Purpose:** Avoid zero probabilities for unseen configurations while controlling parent-set complexity.

Bayesian Network: Structure and Parameters



Exact Probabilistic Inference

- **Inference Method:** Exact variable elimination on the Bayesian Network.
- **Procedure:**
  a. Incorporate the observed evidence.
  b. Multiply the relevant local factors.
  c. Sum out non-query, non-evidence variables.
  d. Normalize the remaining factor to obtain the posterior.
- **Primary Query:** $P(\mathrm{LUNG_CANCER} = 1 \mid e)$
- **Validation:**
  a. Test multiple partial-evidence configurations.
  b. Compare posterior probabilities when evidence variables change.
  c. Validate against brute-force enumeration on small configurations.
  d. Record numerical validation error and inference latency.
- **Requirement:** Report numerical posterior values only from executed inference outputs, with the corresponding evidence configuration.

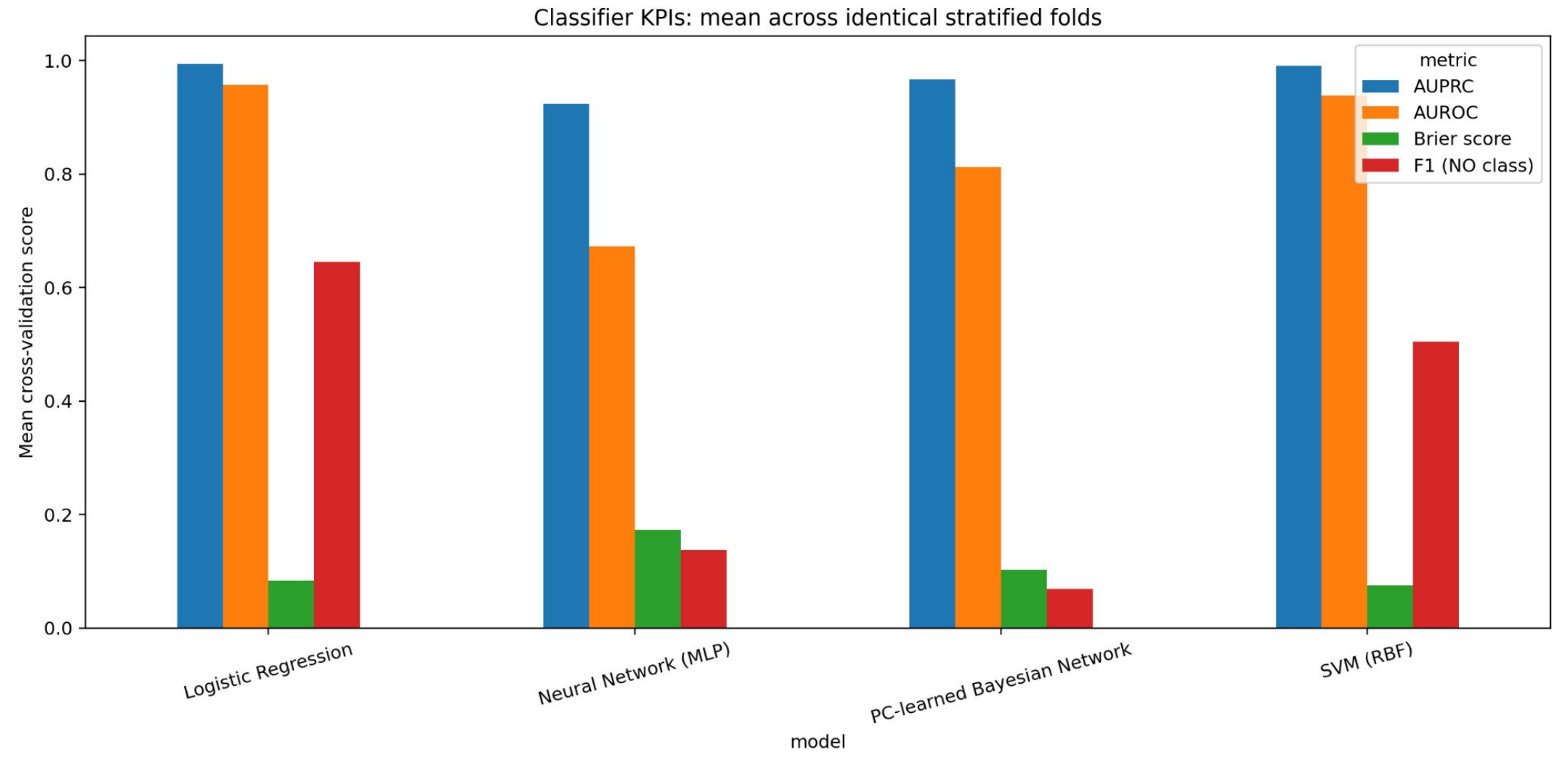
Classification Baselines and Evaluation KPIs

- **Models:**
  - Logistic Regression
  - Neural Network (MLP)
  - RBF-kernel Support-Vector Machine
  - PC-learned Bayesian Network

- **Evaluation Protocol:**
  - Stratified cross-validation.
  - Identical fold assignments across predictive models.
  - Fit preprocessing and probability calibration within training folds.
  - Report fold-level results and mean ± standard deviation.

- **Primary Metrics:**
  - **AUROC:** Threshold-independent ranking quality.
  - **AUPRC:** Positive-class retrieval quality under class imbalance.
  - **F1 (NO class):** Minority-class detection.
  - **Brier score:** Mean squared error of predicted probabilities; lower is better.

Preliminary Classification Results



Explainability Through Bayesian Inference

Discriminative Prediction vs. Probabilistic Inference

- **Procedure:**
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Limitations

- Small sample size and class imbalance.
- Potential unreliability of asymptotic chi-square tests with sparse contingency tables.
- Sensitivity of learned structure to the significance threshold and conditioning-set choices.
- Unresolved CPDAG edge orientations require explicit treatment before DAG-based inference.
- Observational associations do not establish causal effects.
- The survey dataset is not a representative clinical cohort.

Required Validation

- Repeat PC learning at $\alpha \in 0.01, 0.05, 0.10$.
- Inspect contingency tables and expected cell counts.
- Compare edge stability across resampled datasets.
- Validate orientation rules and DAG acyclicity.
- Evaluate BN predictions on held-out folds using fold-specific structure and parameter learning.
- Archive separating sets, graph outputs, fold assignments, predictions, seeds, and experiment scripts.

Conclusion and Research Position

- **Conclusion:**
  - Learn the conditional-independence structure using the PC algorithm.
  - Learn a Bayesian Network (BN) parameterization for probabilistic queries.
  - Compare BN predictions with a neural network and an RBF-kernel SVM.
  - Evaluate classification using AUROC, AUPRC, minority-class F1, and Brier score.
  - Demonstrate explainability through posterior inference under partial evidence.

- **Research Position:**
  - Neural Graphical Models (Shrivastava and Chajewska, 2022) motivate graph-structured neural parameterizations. The present implementation evaluates classical graphical-model and classifier baselines; it does not claim to reproduce the full NGM method.

Team Responsibilities

Member	Primary Responsibility
Priyam Modi	PC Algorithm Basis, Representation, Inference and Learning Design, Programming on the basis of PC Algorithm, and KPI design.
Rushin Modi	Problem framing, Dataset and Preprocessing, Baseline and Programming, Deeper Understanding and Setup of Timeline.
Jash Odedara	Bayesian and Markov Network Differences, and understanding how PC algorithm works; KPI computation and metrics logging; and Repository Structure.

References

D. Colombo and M. H. Maathuis, “Order-independent constraint-based causal structure learning,” JMLR, vol. 15, pp. 3741–3782, 2014.

H. Shrivastava and U. Chajewska, “Neural Graphical Models,” Oct. 2022. arxiv.org/abs/2210.00453v4

N. Alaswad, “Lung Cancer Dataset,” Kaggle.kaggle.com/datasets/nancyalaswad90/lung-cancer

P. Spirtes, C. Glymour, and R. Scheines, Causation, Prediction, and Search, 2nd ed., MIT Press, 2000.

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